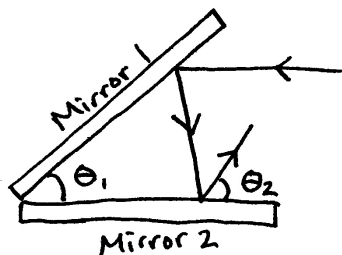


Homework 6: Light and Lenses

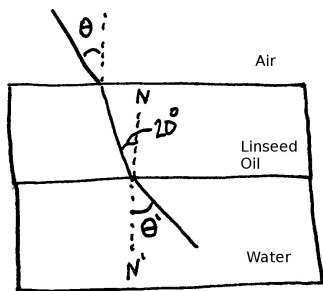
Problem 6.1

Two plane mirrors are at an angle of $\theta_1 = 50.0^\circ$ with each other as in the side view shown in the below figure. If a horizontal ray is incident on mirror 1, at what angle θ_2 does the outgoing reflected ray make with the surface of mirror 2?



Problem 6.2

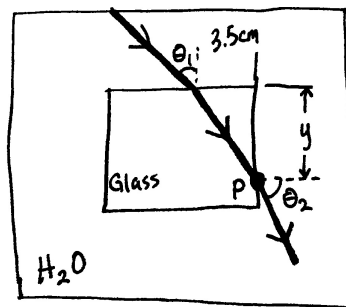
The light beam shown in the below figure makes an angle of 20.0° with the normal line NN' in the linseed oil. Determine the angles θ and θ' . (The refractive index for linseed oil is 1.48.)



Problem 6.3

A block of crown glass is immersed in water as in the below figure. A light ray is incident on the top face at an angle of $\theta_1 = 42.0^\circ$ with the normal and exits the block at point P .

1. Find the vertical distance y from the top of the block to point P .
2. Find the angle of refraction θ_2 of the light ray leaving the block at point P .



Problem 6.4

A beam of laser light with wavelength 612 nm is directed through a slab of glass having index of refraction 1.78.

1. For what minimum incident angle would a ray of light undergo total internal reflection?
2. If a layer of water is placed over the glass, what is the minimum angle of incidence on the glass-water interface that will result in total internal reflection at the water-air interface?
3. Does the thickness of the water layer or glass affect the result?
4. Does the index of refraction of the intervening layer affect the result?

Problem 6.5

A diverging lens has a focal length of magnitude 20.0 cm . An object is placed at distances of (i) 40.0 cm (ii) 20.0 cm (iii) 10.0 cm . For each case:

1. Find the image distance.
2. Find if the image is real or virtual.
3. Find if the image is inverted or upright.
4. Find the magnification.

Problem 6.6

A transparent photographic slide is placed in front of a converging lens with a focal length of 2.44 cm . An image of the slide is formed 12.9 cm from the slide. How far is the lens from the slide if the image is

1. Real?
2. Virtual?

Problem 6.6

Two converging lenses, each of focal length 15.0 cm , are placed 40.0 cm apart, and an object is placed 30.0 cm in front of the first lens. Where is the final image formed, and what is the magnification of the system?

Problem 6.7

Two converging lenses having focal lengths of $f_1 = 10.0\text{ cm}$ and $f_2 = 20.0\text{ cm}$ are placed $d = 50.0\text{ cm}$ apart, as shown in the below figure. The final image is to be located between the lenses, at the position $x = 31.0\text{ cm}$.

1. How far to the left of the first lens should the object be positioned?
2. What is the overall magnification of the system?
3. Is the final image upright or inverted?

